

Feolu Kolawole

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EDUCATION

Stanford University

Expected Graduation Date: June 2028

Pursuing a Bachelor's in Computer Science (AI Track and sub-focus in Human-Computer Interaction)

Completed: CS106B: Data Structures & Algorithms, CS107: Computer Org. & Systems, CS111: OS's & Concurrency, MATH51: Linear Algebra & Multivariable Calculus, MATH104: Applied Matrix Theory, CS231N: Deep Learning for Computer Vision

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C#, C++, Swift, Assembly, HTML/CSS, R

Libraries/Frameworks: Tensorflow, Pytorch, SAM, YOLO, SwiftUI, RealityKit, Bootstrap, Dialogflow, React, Numpy, Pandas

Software: Blender, Unity, Adobe Photoshop

PROFESSIONAL EXPERIENCE

Machine Learning and Computer Vision Researcher - HairFlow

Stanford, CA: March 2025 - Present

Stanford Artificial Intelligence Laboratory (PI - Ron Fedkiw)

- Developed computer vision pipelines for 3D modeling and segmentation, with a focus on enabling human mesh component creation for industry applications in 3D animation, film, and media.
- Engineered models for 3D character and avatar development via hair type classification, extraction, and generation.

Research Assistant

Palo Alto, CA: April 2025 - Present

Stanford Human Perception Lab (PI - Khizer Khaderi)

- Enhanced computer vision pipelines by integrating existing SLAM and Point Cloud Segmentation models to transform video data into accurate 3D environmental models at a rate of approximately 72 frames per second.
- Facilitated advanced simulations and analyses by seamlessly transferring generated 3D models to NVIDIA Omniverse.
- Built a specialized AR application for Snap Spectacles to test users' peripheral vision recognition and reaction time.

PROJECTS

Accelerated MRI Reconstruction with SwinUNet Deep Learning

Python | Pytorch | Deep Learning

Transformer based Deep Learning Architecture for Medical Imaging ([View Paper](#))

- Enabled faster MRI scans for patients by reconstructing high-quality knee MRI images, reducing patient time in the scanner.
- Developed an optimized SwinUNet deep learning architecture for accelerated MRI reconstruction, achieving 33.1 dB PSNR and 0.72 SSIM on the fastMRI dataset.
- Outperformed baseline U-Net by ~4 dB in PSNR, enhancing MRI image quality and detail preservation from undersampled data.

MIT Virtual Reality Hackathon (Healing Heroes) (Winner)

Swift | RealityKit

Innovative VR-based Healthcare Improvement Simulation

- First place in MIT's VR competition where projects were judged on innovation, impact, and technical complexity.
- Designed & Engineered a VR healthcare simulation on the Apple Vision Pro, allowing users to practice CPR & first aid.

Music Recommendation System ([View Paper](#))

Python | SVD | Dimensionality Reduction | PCA

- Designed a novel music recommendation system utilizing Singular Value Decomposition (SVD) and Principal Component Analysis (PCA) for dimensionality reduction of Spotify audio features.
- Implemented feature projection to identify non-obvious musical similarities, focusing on sound qualities over artist names.

SynchroSound

SwiftUI | UIKit | SwiftData | Objective-C

iOS App for Song Recommendations Based on Facial Mood Recognition

- Built an app that enhances a user's musical experience - deciphers facial expressions and recommends mood-matching songs by integrating SwiftUI, UIKit, and SwiftData with Google Cloud Vision's facial recognition and Spotify's Web API

Supreme Court Case Outcome Prediction Using Bayesian Networks

Python | Numpy | Pandas

Bayesian Model for Predicting Supreme Court Case Dispositions ([View Paper](#))

- Structured a Bayesian network to predict Supreme Court case dispositions based on observable case attributes, creating a probabilistic framework for judicial decision modeling.
- Implemented rejection sampling for inference, achieving a Top-3 accuracy of 73% for prediction amongst 11 possibilities.
- Awarded class winner for Stanford CS109 out of ~500 students, demonstrating excellence in probabilistic modeling.

LEADERSHIP EXPERIENCE

Stanford XR Vice President Of External Affairs

Stanford, CA: September 2024 - Present

- Co-led successful launch of partnerships with prominent sponsors including Meta, NVIDIA, Amazon, and Snapchat
- Organized immersive technology hackathons with a team, engaging over 300 participants in XR and Spatial Computing projects